

Project Proposal - All Digital PLL Synchronizer

ECE298A - TinyTapeout Project

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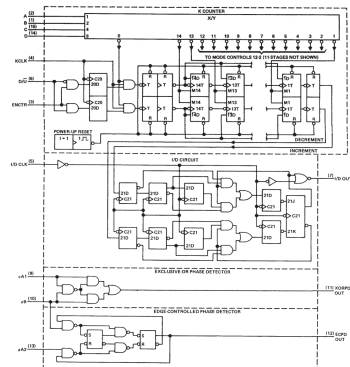
1. Project Overview

- a. What is a digital PLL Synchronizer?
 - A digital PLL (Phase Locked Loop) synchronizer is a clock that aligns the phase of a clock with a reference clock.
 - They're often used in communication receivers and (our application) chip deskewing
- b. Usefulness?
 - This will ensure on chip timing is reliable
 - Can be fabbed as a reusable IP block

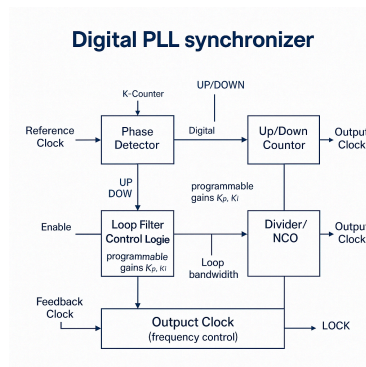
2. Objectives

- a. Implement first order digital PLL Synchronizer using Verilog
- b. Verify functionality with CocoTB python scripts
- c. Synthesize for TinyTapeout
- d. Demonstrate created PLL can lock onto a reference signal, maintain phase alignment, read fuzzy signals and produced realigned phases

3. Block Diagram



[1]



[3]

4. Project Timeline

- a. Week 3 (Sept 17): Built/tested counter in Verilog.
- b. Week 4 (Sept 24): Proposal submitted; counter working.
- c. Weeks 5–6 (Oct 1–8): Implement and test phase detector + divider.
- d. Week 8 (Oct 22): Sub-block evaluation.
- e. Weeks 9–10 (Oct 29–Nov 5): Full PLL integration, synthesis with OpenLane.
- f. Week 11 (Nov 12): System integration milestone.

- g. Weeks 12–13 (Nov 19–26): Final verification + documentation.
- h. Dec 3: Submission.

5. Design Details

- a. *Counter*: Up/Down counter coded in Verilog and tested with CocoTB - Done
- b. *Phase Detector*: Output Up/Down pulses depending on lead/lag of input
- c. *Loop filter*: Accumulates corrections to smooth jitter of incoming signal
- d. *Divider/NCO*: Generates controlled clock output
- e. *Lock detector*: Monitors stable phase alignment

6. Tools and Verification

- a. *Coding*: Verilog will be used (Synthesizable to TinyTapeout)
- b. *Verification*: CocoTB testbenches in Python
- c. *Synthesis & Layout*: OpenLane
- d. *Version control*: Github actions for CI

7. Expected Outcome

- a. A working PLL synchronizer block fitting within 1 TinyTapeout file
- b. *Demo*: PLL locks onto a reference clock in simulation
- c. Ready for Fab via TinyTapeout

8. References

[1] Texas Instruments, *SN54/74LS297 Digital Phase-Locked-Loop Filters Datasheet*, SDLS155, Rev. March 1988. [Online]. Available: <https://www.ti.com/lit/ds/symlink/sn74ls297.pdf>

[2] Texas Instruments, *Digital Phase-Locked Loop Design Using SN54/74LS297*, Application Report SDLA005B, Mar. 1997. [Online]. Available: <https://www.ti.com/lit/an/sdla005b/sdla005b.pdf>

[3] Artificially generated image by ChatGPT after receiving the following prompt: “Can you create a low-level block diagram of a digital PLL synchronizer that is to be made on a tile, on a chip”